

LED FLASHER

This is regarding the DIY article ‘An Inexpensive 5-Colour LED Flasher’ published in April issue. May I ask what is the purpose of the 1N4148 diodes used in the circuit? If these are for reversed current, I would think that both outputs Q1 and Q2 (and the others) should be connected to the anodes of the diodes. Please confirm whether I am correct?

Tintintin

EFY. We thank the reader for detecting this error and sharing with everyone! Please refer the circuit diagram and connect anodes of 1N4148 diodes to the outputs of Q1, Q3, Q5, Q7, and Q9—just like the connections to Q0, Q2, Q4, Q6, and Q8—and the cathodes of 1N4148 diodes to the junctions before the base bias resistors of respective transistors. Yes, these diodes are for preventing reversed current.

• • •

APPLIANCES’ PROTECTION

This is regarding the DIY article ‘Over-voltage and Undervoltage Protection for Electrical Appliances’ published in April issue. What is the need for two 1N4007 diodes in parallel when the transformer itself is rated only for 500mA, and when there is not much load? Please provide a clarification.

V. Ramprakash

The author Shamail Ahmad replies: The transistor T1 (BC337) can handle up to 800mA current, so there is flexibility in the circuit to drive much higher load relays or even contactors. Also, the transformers sold in the local market are generally overrated. The 38mm × 32mm core transformer used in the circuit is typically rated at 500mA, but the actual current that comes out is barely 200mA.

EFY. The 1N4007 diodes (D2 and D3) in parallel are only precautionary (to support the other two diodes). The

circuit’s working will not be affected even if you remove these two diodes.

• • •

CHATGPT

This is regarding the DIY Cool Stuff article ‘How to Convert ChatGPT into an Advanced Voice Assistant’ published in March issue. This is a very informative article, but can we use it for different AI languages?

Chithraa

EFY. Yes, you can do that but for that you shall have to make changes in the code and use the NLP model for that language.

• • •

INVERTER THD

What is the THD of output in an inverter and how to minimise it?

Mitesh

EFY. THD stands for Total Harmonic Distortion, and it is a measure of the amount of harmonic distortion present in a signal compared to its original frequency. It can be calculated for various signals, such as audio signals, power signals, or any other periodic signal.

In an inverter, the THD of the output signal can vary depending on the type of inverter and the load being driven by the inverter. In general, the output waveform of an inverter is a pulse width modulated (PWM) signal, which is a series of low- and high-voltage pulses within the frequency range of the inverter.

The THD of the output waveform can be affected by various factors, such as the switching frequency of the inverter, the shape of the output waveform, and the type of load being driven. In general, the THD of an inverter output waveform should be as low as possible to ensure proper operation of the load and to minimise electromagnetic interference.

To measure the THD of an inverter

output waveform, a distortion analyser or a spectrum analyser can be used. The THD is typically expressed as a percentage and is calculated by measuring the total harmonic content of the waveform and dividing it by the amplitude of the fundamental frequency. A low THD value indicates a clean output waveform with little distortion, while a high THD value indicates a waveform with significant harmonic distortion.

There are several ways to minimise the THD of an inverter output waveform, such as:

Increase the switching frequency.

Higher switching frequencies result in smaller voltage steps, which reduces the amount of harmonic distortion in the output waveform.

Improve the waveform quality.

The output waveform can be improved by using advanced pulse width modulation (PWM) techniques, such as sinusoidal PWM, third harmonic injection PWM, or space vector PWM.

Use a filter. A filter can be added to the output of the inverter to reduce the amount of harmonic distortion. The most common filter used for this purpose is a low-pass filter, which attenuates the high-frequency harmonic components of the waveform.

Match the load. The load being driven by the inverter can affect the THD of the output waveform. Matching the load to the inverter’s output characteristics can reduce the amount of harmonic distortion.

Use high-quality components.

Using high-quality capacitors and inductors can reduce the amount of harmonic distortion in the output waveform.

By implementing one or more of above techniques, the THD of an inverter output waveform can be minimised, resulting in a cleaner output signal with reduced electromagnetic interference.

Letters and questions for publication may be addressed to: Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: onedit@efy.in)